Data Preprocessing Report - Feature Engineering & Data Preprocessing Prepare

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Data Cleaning

Columns with Null Values	
Gender	13
Married	3
Dependents	15
Self_Employed	32
LoanAmount	22
Loan_Amount_Term	14
Credit_History	50

Handling Missing values

Missing values were replaced with either the mode or the median of the values in the column dependent on which was of great use. Mode was used for all columns except Loan Amount Median was used for this column because Loan Amount may have bigger numbers. The relevant data types were changed as well, Gender, Dependents, Self Employed were changed to str(strings). The Loan Amount Term and Credit History were changed to interger. There were no duplicates in the set.

Renaming

Columns Property Area, Loan Amount Term, Education and credit History were changed to be Area Type because the values speak to different areas such as rural, semi urban hence Area Type is best suited. Loan Amount Term to Loan Duration Months which explains the values in the columns that the duration is in months. Education changed to graduate because the values there are either Graduated or Not and Credit History to Credit Score. Loan ID column was dropped having no relevance for the predictive model.

Feature Encoding Apply suitable encoding techniques such as:

We will be using ONE Label encoding for columns Gender, Married, Self-Employed, Graduated and Loan Status. These columns contain categorical values hence ONE label will be best suited for them. We will then use ONE HOT Encoding for the different Area Types as they need to be binary

Feature Scaling

Detecting and Handling Outliers

We will use Boxplots to check the nature of outliers in the different columns after which we will use the IQR method to handle these outliers, the IQR method is best suited for outliers, and it essentially converts the outlier to either the lower or upper quantile making it easier for our model to predict future variables.

Standardization

Robust Scaler is useful for these columns or variables because these variables have varying datapoints and they may have a lot of outliers hence robust is the best scaler for these variables or features. Standard Scaler is best suited for Loan Duration Months.

Feature Selection Identify:

Checking correlation

Correlation will help us dictate which variables are worth using for Machine learning and which do not correlate with Loan Status

Features which would work best for Machine learning and features which wont work Top 6 best feautures would be

- Credit Score which is 0.540
- Area Type Semiurban 0.136
- Married 0.096
- Gender 0.018
- Coapplicant Income 0.012
- Dependents 0.010

Since Applicant Income, Coapplicant Income and Total Income are vital for business to make a decision on the final outcome we shall include it because it is very relevant for the business predictive loan model we trying to do.

Total Income shall be removed because total Income is included in both the income columns